# MAROKA MUSA CALVIN

📧 calvinmaroka@gmail.com | 📱 (+27) 63 607 1420

- 🔗 LinkedIn: Maroka Musa | 🔗 GitHub: github.com/Calvin527 | 🔗 Portfolio: *Live link*

## PROFESSIONAL SUMMARY

### SOFTWARE ENGINEER | FULL-STACK DEVELOPER

Junior Software Engineer and Full-Stack Developer with hands-on experience building REST APIs, database-driven applications, and real-world software systems. Experienced in software engineering practices, application testing, debugging, database design, and backend development. Skilled in Python, JavaScript, Flask, Django, Node.js, PostgreSQL, Git, and GitHub, with additional exposure to machine learning and AI. Focused on building scalable, reliable, and user-focused software solutions.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C#

**Backend Development:** Node.js, Flask, Django REST Framework, REST API, JWT Authentication

**Frontend Development:** React.js, HTML5, CSS3, Vite.

**Databases:** PostgreSQL, SQL

**Software Engineering:** Requirements Engineering, UML Modelling, Object-Oriented Design, Software Testing, Debugging.

**Machine Learning & Data Science:** Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib.

**Systems & Networking:** Linux, Operating Systems, Computer Networking

**Tools:** Git, GitHub, VS Code, Render, Gunicorn, Linux.

## PROFESSIONAL CERTIFICATION

**Aug 2026 | Python Django for Beginners — Edureka / Coursera** | **credential**

- Built practical knowledge of Django project and app architecture, URL routing, views, templates, and reusable components.
- Worked with Django models, migrations, ORM, forms, ModelForms, and CRUD operations.
- Implemented user authentication, access control, media handling, and dynamic database-driven pages.
- Learned production preparation concepts including static files, security settings, Gunicorn, and deployment workflows.
- Applied Django concepts to strengthen practical Software Engineering project development.

**Sept 2026 | Software Engineering: Modeling Software Systems using UML — The Hong Kong University of Science and Technology / Coursera |** *credential*

• Built practical knowledge of the Software Development Life Cycle (SDLC), including planning, requirements analysis, design, implementation, testing, and maintenance.

• Learned how to use UML models to capture system requirements and improve communication between users, developers, and stakeholders.

• Worked with use-case modelling to describe functional requirements and interactions between users and the system.

• Gained understanding of domain modelling, including classes, associations, constraints, and important application-domain concepts.

• Applied object-oriented modelling principles to strengthen software design, system documentation, and software engineering project development.

**Aug 2026 | Python for Data Science, AI & Development – IBM** | ***credential***

- Developed practical Python skills using core data structures and programming concepts for data-driven applications.
- Worked with Pandas and NumPy for data manipulation, cleaning, wrangling, and analysis.
- Used REST APIs and HTTP requests to retrieve and process external data.
- Practiced data collection and web scraping using BeautifulSoup and Selenium.
- Strengthened practical foundations for data science, machine learning, and backend development.

## PROJECTS

**October 2026 | TAXI MOVE — Smart Taxi Demand Monitoring and Dispatch System | Live Demo | GitHub**

- Applied software engineering principles across the SDLC, including requirements engineering, system analysis, object-oriented modelling, software design, implementation, testing, and deployment.
- Conducted requirements analysis and developed software specifications and UML models to represent system functionality, user interactions, class relationships, and object-oriented design.
- Engineered a full-stack application using React.js and Django REST Framework, implementing role-based authentication, REST APIs, GPS-based passenger check-ins, geofenced waiting spots, and taxi dispatch management.
- Designed relational database models using PostgreSQL, incorporating data relationships, validation constraints, and business rules for passenger requests, driver availability, routes, and trip management.
- Applied software testing, debugging, and version control practices to improve system reliability; deployed the frontend and backend on Render, integrating Neon PostgreSQL for persistent cloud data storage.

**Tech:** Python, Django REST Framework, React.js, JavaScript, PostgreSQL, Neon, REST APIs, JWT, Leaflet, Git, GitHub, Render.

**Apr 2025 | Makgobelo Lodge Booking System** | *Live Demo*

Developed a full-stack B&B reservation management system with real-world functionality.

• Implemented secure authentication with email verification and role-based access (Admin/User)
• Developed booking system with date validation to prevent double-booking

- Built admin dashboard for managing users, rooms, and booking history
- Integrated email notifications for verification, bookings, and cancellations
- Designed and implemented relational database structure

**Tech:** Python (Flask), Neon PostgreSQL, HTML, CSS

**JUN 2024 | MAROKA REAL-TIME MEETING DISCUSSION | *Live Demo***

Developed a real-time meeting discussion platform that supports live communication, saved chat records, user mentions, replies, online user tracking, and meeting recap downloads.

- Implemented real-time messaging using Socket.IO and WebSockets
- Added user join/leave tracking, online user lists, and duplicate-name prevention
- Built saved discussion records for chat history, decisions, tasks, and shared files
- Developed features for mentions, replies, emoji reactions, voice notes, and file sharing
- Prepared the application for deployment using Node.js and Express

**Tech:** Node.js, Express.js, Socket.IO, JavaScript, HTML, CSS, JSON Storage

**Mar 2025 | Cloud Operating System Simulation | *GitHub***

Simulated backend system behaviour for cloud resource management.

- Developed algorithms for dynamic resource allocation
- Implemented concurrency control using synchronization techniques
- Simulated system performance under varying workloads
- Analysed system metrics including CPU and memory usage

**Tech:** C# (.NET), PostgreSQL

## CERTIFICATIONS

**Python Programming for AWS: Learn Python with AWS and Boto3 — Udemy, 2026 | *credential***

Completed training in using Python with AWS cloud services through Boto3. The course helped me understand how Python can be used to automate cloud tasks, interact with AWS services, and support backend/cloud-based applications.

**Python on the Backend — Udemy, 2026 | *credential***

Completed a backend-focused Python course that improved my understanding of server-side development. The course covered how Python is used to build backend systems, process requests, handle application logic, and support web-based applications.

## EDUCATION

**BSc in Mathematics and Computer Science** | University of Limpopo

2023 – 2025

**Honours Degree in Computer Science** | University of Limpopo

Jan 2026 – Nov 2026

## KEY STRENGTHS

- Software engineering, requirements analysis, UML modelling, and object-oriented system design.
- Back-end development, REST API design, and relational database management.
- Full-stack application development using Python, Django REST Framework, React.js, and PostgreSQL.
- Software testing, debugging, Git/GitHub version control, and cloud deployment.
- Machine learning, data preprocessing, and data analysis using Python and modern ML frameworks.
- Analytical problem-solving, systems thinking, teamwork, and adaptability to new technologies.